

Sambhav Koshta

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B.Tech - Computer Science and Engineering

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LeetCode Codeforces CodeChef

EDUCATION

Maulana Azad National Institute of Technology, Bhopal

Aug 2023 – present

B.Tech Computer Science and Engineering

CGPA: 8.18

Royal Senior Secondary School, Jabalpur

May 2023

Class XII, CBSE

Percentage: 88

Joy Senior Secondary School, Jabalpur

May 2021

Class X, CBSE

Percentage: 91

PERSONAL PROJECTS

TurfMate — Sports Turf Booking Platform

 Live  GitHub Aug 2025

React.js · Node.js · Express.js · MongoDB · Mongoose · JWT · Cloudinary · Vercel

- Built **tri-role full-stack platform** (Player/Owner/Admin) with **JWT + bcrypt authentication** and role-scoped middleware across **30+ REST endpoints** on **Node.js, Express, and MongoDB Atlas**.
- Designed **conflict-safe slot reservation engine** with real-time availability checks, **5-minute confirmation window with auto-expiry**, and concurrent booking protection using **Mongoose atomic operations**.
- Implemented **multi-tenant revenue management** with automated commission splitting, wallet ledgering, and owner earnings analytics dashboard — handling full **booking lifecycle from creation to payout**.
- Deployed **dual React 18 SPAs** (player + admin panel) on **Vercel CDN** with **Cloudinary media pipeline** for turf listings, document verification uploads, and image optimization.

Identra AI — AI-Powered eBook Creation Platform

 Live  GitHub Dec 2025

React.js · Node.js · Express.js · MongoDB · JWT · Gemini AI · PDFKit · Render

- Engineered **AI content generation pipeline** using **Google Gemini API** to produce structured book outlines and chapter content from user prompts, integrated into a **real-time markdown editor with live preview**.
- Built **drag-and-drop chapter management system** using **DnD Kit** with persistent chapter ordering in **MongoDB**, alongside **multi-format export** (PDF via PDFKit, DOCX) with one-click download.
- Implemented **JWT-based authentication** with **bcrypt** password hashing, **Multer** file upload handling for cover images, and per-user private book library with full **CRUD operations**.
- Deployed full-stack application on **Render** with Express serving the **Vite-optimized React build** in production, configured with **environment-based CORS** and secure HTTP-only cookie auth.

App Data Collection Predictor — ML Privacy Analysis Tool

 Live  GitHub Jan 2026

Python · Flask · scikit-learn · Random Forest · Decision Tree · pandas · NumPy · Chart.js

- Built **multi-label classification system** using **Random Forest** and **Decision Tree** to simultaneously predict **4 privacy data-collection categories** across 200 Google Play Store apps — achieving **weighted F1-score of 0.85** with **class-imbalance handling** via balanced_subsample weighting for a highly skewed Contacts label (3.5% positive rate).
- Engineered **leakage-free feature extraction pipeline** deriving **8 behavioral signals** from raw app metadata JSON — maintaining strict separation between feature and label derivation across **80/20 train-test split** with fixed random state for reproducibility.
- Deployed **Flask REST API** serving real-time multi-label predictions at ~50ms latency via **MultiOutputClassifier** wrapper, with a **vanilla JS frontend** featuring **Chart.js** model comparison, dynamic app auto-fill, and color-coded prediction cards.

SKILLS

Languages: C, C++, JavaScript, Python

Frontend: React.js, Tailwind CSS, HTML, CSS, Vite

Backend: Node.js, Express.js, Flask, REST APIs, JWT Authentication

Databases: MongoDB, Mongoose, MySQL, PostgreSQL

ML / AI: scikit-learn, pandas, NumPy, Gemini AI, matplotlib

DevOps & Tools: Git, GitHub, Docker, Vercel, Render, Postman, VS Code, Jupyter Notebook

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- Selected among **top teams at institute level** for **Smart India Hackathon (SIH '25)**.
- Secured **District Top rank** in **Technothon** conducted by IIT Guwahati.
- Solved **500+ DSA problems** across LeetCode and GeeksforGeeks.
- Rated **1250 (Pupil)** on Codeforces and achieved **2-star (1433)** on CodeChef.